songgot · 송곳

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A Korean-first tiny agentic model, for tool calling on the device.

Songgot is a from-scratch tiny language model (tens of millions of parameters) for Korean tool calling and structured extraction on phones and small devices, with a Korean-first 32k tokenizer, licence-clean data, and reproducible exact-match evaluation on Kakao FunctionChat-Bench. Weights, code and paper under Apache 2.0.

Weights on Hugging Face | Code on GitHub | Live demo | Paper as PDF | Paper as Markdown

0.90
tokens per Hangul syllable (Songgot 32k tokenizer); Needle 2: 3.47

500
Korean items, FunctionChat-Bench SingleCall, exact-match scorer

Apache 2.0
weights, tokenizer, code, data generators, paper

0 closed-model labels
licence-clean data, provenance in the paper

Hanish Keloth, Palette (Seoul and Bengaluru). Draft of 2026-09-10. Numbers marked TBD are filled from the run logs before release; nothing here is estimated.

Abstract

Tiny agentic models (Needle 2, 45M parameters, 14 MB, July 2026) made on-device tool calling practical on phones and microcontrollers, but the released model is English only. Songgot is a Korean-first model in the same class: a decoder trained from scratch with a Korean-first 32k tokenizer, bilingual Korean and English pretraining on licence-clean text, and post-training on Korean tool-calling and structured-extraction data that no closed model touched. We evaluate on Kakao's FunctionChat-Bench SingleCall (500 Korean items) with a deterministic exact-match scorer that anyone can reproduce without an API key, against Needle 2, FunctionGemma-270M and Qwen3-0.6B under identical prompts. Results: TBD. Weights, tokenizer, data generator, scorer and this paper are released under Apache 2.0.

1. Why a Korean-first tiny model

- The class exists and is growing: Needle 2 passed 52,000 downloads within weeks of release.
- On Hugging Face as of 2026-09-09 there is no Korean tool-calling model under 1B parameters; the one Korean function-calling finetune found is a 2B Gemma.
- Korean is expensive for English-first tokenizers. On the 100 FunctionChat SingleCall queries, Qwen3's 151k vocabulary spends 1.15 tokens per Hangul syllable, Gemma's 262k vocabulary 0.98, and Songgot's 32k Korean-first vocabulary 0.90 (Table 1). At a 256 to 1024 token budget shared with tool schemas, that is context.

2. Model

Llama-style decoder: TBD layers, hidden 512, GQA with 8 query and 2 key-value heads, SwiGLU intermediate 1408, RoPE, tied embeddings, 32k vocabulary, TBD M parameters. Trained from scratch. Exports to safetensors, GGUF (f16, Q8_0, Q4_K_M) and runs under llama.cpp.

Figure 3. Songgot-nano pretraining loss on the Mac, read from the run log

3. Tokenizer

SentencePiece BPE, 32,000 pieces, byte fallback, digits split, trained on a 300 MB sample: Korean Wikipedia 50 percent, fineweb-edu 40 percent, tool-schema JSON 10 percent. Special tokens <|system|> <|user|> <|call|> <|end|> <|pad|>.

Table 1. Tokens per Hangul syllable on the 100 FunctionChat SingleCall queries (2,071 syllables).

TOKENIZER	VOCAB	TOKENS	TOKENS PER SYLLABLE
Songgot (ours)	32k	1,873	0.90
Gemma 3 / FunctionGemma	262k	2,028	0.98
Qwen3	151k	2,392	1.15
Needle 2	8k	7,192	3.47

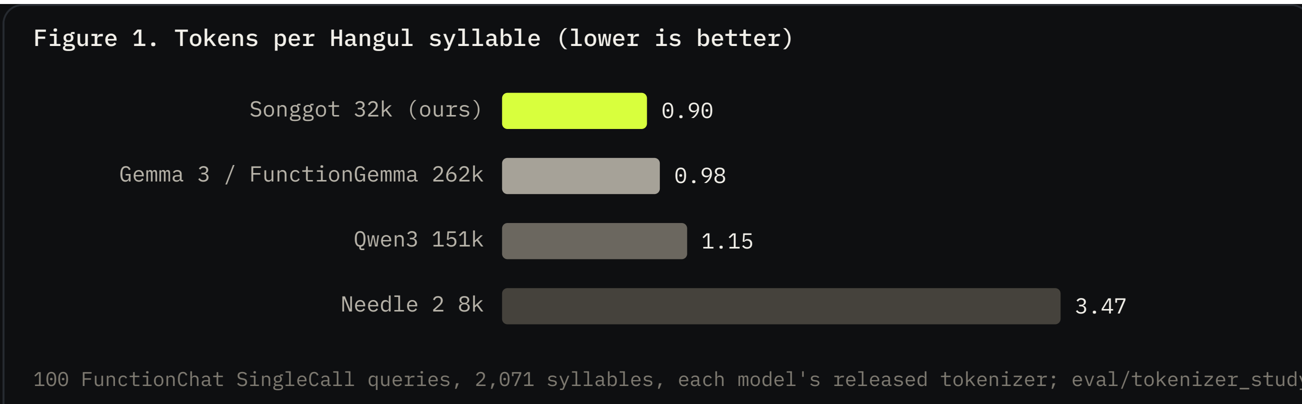


Figure 1. Tokens per Hangul syllable on the FunctionChat queries, lower is better

4. Data

Pretraining (all disclosed, all licence-clean):

- fineweb-edu sample-10BT (ODC-By), first 1.63B tokens under our tokenizer.
- Korean Wikipedia, dump 20231101.ko (CC BY-SA 3.0), 0.60B tokens, upsampled to roughly half of the training mix.
- No AI-Hub data (its terms forbid leaving Korea and our GPUs do not sit there), no crawled Korean web text of unclear licence.

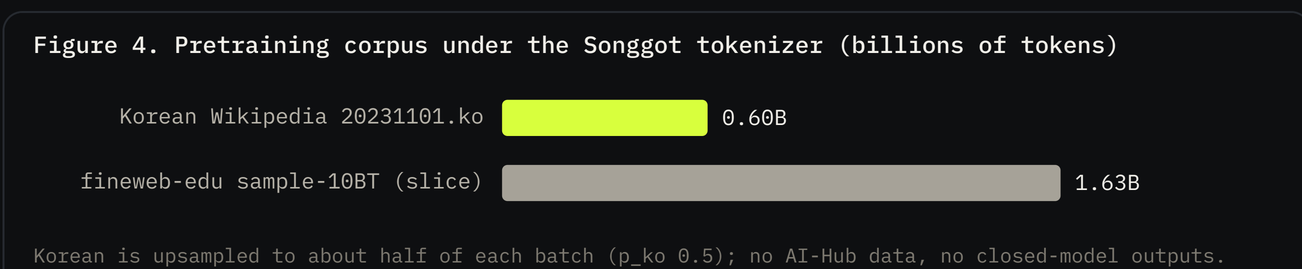


Figure 4. Pretraining corpus in tokens under the Songgot tokenizer

Post-training (85,408 examples): 64,000 Korean tool calls generated from hand-written frames for 58 Korean tools (messaging, calendar, transit, home appliances, commerce, public services, health, work, information), with register variants (반말, 존댓말, 격식, 사투리, typos), Korean date and time words resolved against a fixed calendar, and 4,000 "no tool applies" negatives; 21,408 English examples from glaive-function-calling-v2 (Apache 2.0). No closed model produced a label. A disjointness gate aborts the build if any training tool name or query appears in FunctionChat-Bench; three tool names had to be renamed because of it.

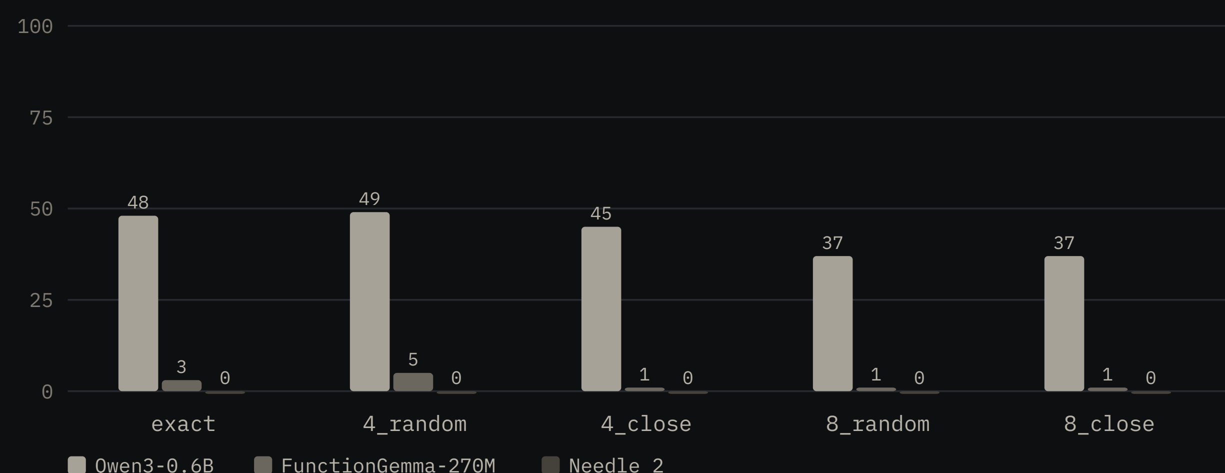
5. Evaluation

FunctionChat-Bench SingleCall (Kakao, 2024, Apache 2.0): 25 functions, 4 Korean queries each, 5 tool conditions (exact, 4 random, 4 close, 8 random, 8 close) = 500 items. The official protocol uses GPT-4 as judge; we score with exact match on function name and arguments (numbers compared as numbers, acceptable alternatives honoured) so the number is reproducible offline. Every model receives the same tools and query, rendered in its own documented format.

Table 2. Call accuracy (exact match) on SingleCall, by tool condition. TBD from run logs.

MODEL	PARAMS	EXACT	4_RANDOM	4_CLOSE	8_RANDOM	8_CLOSE	ALL	NA OI
Songgot	TBD							
Needle 2	45M	0.0	0.0	0.0	0.0	0.0	0.0	0.
FunctionGemma-270M	270M	3.0	5.0	1.0	1.0	1.0	2.2	3
Qwen3-0.6B	600M	48.0	49.0	45.0	37.0	37.0	43.2	7

Figure 2. FunctionChat-Bench SingleCall call accuracy by tool condition (percent)



Exact match, 100 items per condition. Needle 2 is English only and answered none.

Figure 2. Call accuracy by tool condition on FunctionChat-Bench SingleCall

6. Honest reading

- TBD after results: where Songgot loses, and why (argument-value paraphrase, rare tools, English-only failures).
- The template-generated Korean data is narrow by construction; a teacher-generated set from our own Palette-K-Midm was planned and is queued behind GPU availability.
- Pretraining tokens are TBD; the Needle 2 recipe used 200B. Ours is smaller and says so.

7. Release

Weights and tokenizer: hf.co/imcapsule/songgot. Code, data generators, scorer, paper: github.com/hanishkeloth/songgot. Licence Apache 2.0. Korean Wikipedia attribution and CC BY-SA notice in the model card.

Questions and answers

▶ **What is Songgot?**

▶ **How big is it?**

▶ **How is it different from Needle 2?**

▶ **What data was used?**

▶ **How is it evaluated?**

▶ **Where are the weights and code?**

About the author

Songgot is built by **Hanish Keloth**, CTO at Palette (Seoul and Bengaluru), where he leads Palette OS, a company operating system in which governed AI agents draft the documents Korean companies run on. Palette publishes its Korean models and benchmarks in the open: Palette-K-Midm (first of seven on PALETTE-BENCH-KO v0.2, 2026-08-21), Palette-K-Doc, Palette-K-Speech and Palette Video, all at hf.co/imcapsule. Songgot continues that line at the smallest possible size.

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